

RESEARCH ARTICLE

Effect of digital technology interventions for cognitive function improvement in mild cognitive impairment and dementia: A systematic review and meta-analysis

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Abstract

The development and commercialization of digital therapeutics are increasing. The aim of this study was to determine the effects of digital technology interventions on cognitive function, thereby providing evidence for the development and practical application of interventions to manage cognitive function in patients with mild cognitive impairment and dementia. We conducted a systematic review and meta-analysis of randomized controlled trials according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines. Randomized controlled trials on digital technology interventions published until April 2023 were searched in PubMed, Embase, Cochrane Library, and Cumulative Index to Nursing and Allied Health Literature (CINAHL) databases without a period limit. Articles that identified the effects of digital technology interventions on cognitive function improvement in dementia and mild cognitive impairment were integrated and analyzed. RevMan software 5.4 was used for quality assessment and meta-analysis. Twelve out of 708 studies were included in the review and meta-analysis. Digital technology interventions had significant effects on global cognitive function (standardized mean difference [SMD] = 0.31; 95% confidence interval [CI] = 0.04–0.57; $p = 0.02$; $I^2 = 60\%$). In addition, these interventions had significant effects on neuropsychological characteristics, including attention (SMD = 1.17; 95% CI = 0.36–1.97; $p = 0.004$; $I^2 = 84\%$), visuospatial perception (SMD = 0.68; 95% CI = 0.19–1.17; $p = 0.006$; $I^2 = 57\%$), and memory (SMD = 0.45; 95% CI = 0.19–0.71; $p = 0.0007$; $I^2 = 22\%$). The results suggest that digital technology interventions help improve cognitive function in patients with dementia and mild cognitive impairment.

KEYWORDS

cognitive dysfunction, dementia, digital technology, meta-analysis

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1 | INTRODUCTION

Dementia refers to a clinical syndrome that affects one's daily life due to deterioration in cognitive functions, such as memory, language, attention, visuospatial, and frontal lobe executive functions (American Psychiatric Association, 2022). According to a recent report by the World Health Organization, the number of patients with dementia worldwide was estimated to be 55 million in 2019 and is expected to increase to 139 million by 2050 (World Health Organization, 2021). Moreover, social costs of dementia accounted for 1.09% of the world's gross domestic product in 2015, and the economic burden is expected to increase owing to an increase in the cost per person with dementia (Alzheimer's Disease International, 2022).

Mild cognitive impairment is a pre-stage of dementia, and every year about 10%–15% of mild cognitive impairment is converted to dementia caused by Alzheimer's disease (Tábuas-Pereira et al., 2016). Approximately 80% of people exhibit symptoms of dementia within 6 years after the onset of mild cognitive impairment (Petersen, 2003). Early diagnosis of mild cognitive impairment and dementia and prompt therapeutic intervention are important to lower the social cost by slowing the rate of cognitive decline. According to projections of an Australian study, delaying the onset of dementia by 6 months and 5 years predicted 5.6% and 44% lower prevalence of dementia, respectively, 50 years later (Jorm et al., 2005). In this context, providing therapeutic intervention at mild cognitive impairment and early dementia stages is crucial to reduce the overall prevalence of dementia.

Cognitive function is used as an index to diagnose mild cognitive impairment and dementia and includes domains of attention, language, visuospatial perception, judgment, calculation, performance, and memory (Mendez & Cummings, 2003). Domains of cognitive function are also used to predict progression to dementia and to evaluate response to treatment (Rozzini et al., 2008).

Various risk factors and diseases that cause dementia have been identified; however, a comprehensive treatment strategy for dementia to alleviate symptoms or slow the progression has not yet been developed (Tisher & Salardini, 2019). Pharmacological and nonpharmacological treatments for improving cognitive function, relieving behavioral and psychological symptoms, and rehabilitation of motor function have been applied in an integrated manner. In particular, various nonpharmacological treatments, such as cognitive stimulation, cognitive function rehabilitation, memory recall, and music and art therapies are used to maintain and improve the cognitive function of patients with dementia. Recently, with the development of information and communication technology, the nonpharmacological treatment of digital health implemented using computer devices, robots, and virtual reality has been introduced (Zucchella et al., 2018).

Digital health concerns the integration of information communication technology with the medical field; it includes all health services provided using digital technology and aims to collect and promote knowledge of design and implementation of digital health interventions (Jandoo, 2020). Digital therapeutics, a part of digital health, are

defined as clinically-evaluated therapeutic interventions driven by high-quality software programs to prevent, manage, or treat medical disorders or diseases. Digital therapeutics are used alone or in combination with drugs, devices, or other regimens to optimize patient care and health outcomes (Digital therapeutics Alliance, 2022). Digital therapeutics are increasingly in demand because of their relatively low cost compared with conventional treatments and the possibility of personalized interventions (Dang et al., 2020). Various digital therapeutics, such as BluStar for diabetes management, reSET for drug addiction treatment, Sleepio for insomnia, and HelloBetter for chronic pain, have been approved by the US Food and Drug Administration (FDA) and are currently in use (Digital Therapeutics Alliance, 2022). However, FDA-approved digital therapeutics for cognitive function in mild cognitive impairment or dementia have not yet been developed.

To develop effective digital therapeutics for mild cognitive impairment and dementia, it is necessary to review the available digital technology interventions. In previous studies analyzing effects of digital technology interventions, the effects of computerized cognitive training that included hardware devices were analyzed (Hill et al., 2017), whereas the effects on cognitive and motor functions were analyzed only for virtual reality-based interventions (Zhong et al., 2021). In addition, most of the cases included Nintendo's Wii, which was not developed to improve cognitive function in people with mild cognitive impairment and dementia (Hill et al., 2017; Zhong et al., 2021). There is a lack of systematic literature review and meta-analysis identifying characteristics in various types of digital technology interventions that can be the basis for the development of digital therapeutics and suggest an effect size for cognitive function. Although the development and provision of effective digital technology interventions to improve and maintain cognitive function in mild cognitive impairment and dementia are recommended, evidence for their application in dementia clinical nursing is lacking. Therefore, this study was conducted with the purpose of identifying the effects of digital technology interventions on cognitive function and thereby providing evidence for the development and practical application of interventions to support and improve cognitive function in patients with mild cognitive impairment and dementia.

The specific purposes were: (1) to identify the characteristics of selected digital technology intervention studies on individuals with mild cognitive impairment and dementia, (2) to examine effects of digital technology interventions on global cognitive function, and (3) to examine effects of digital technology interventions on neuropsychological characteristics such as attention, visuospatial perception, memory, and executive function.

2 | MATERIALS AND METHODS

This was a systematic literature review and meta-analysis of randomized controlled trials to identify the effects of digital technology interventions for cognitive function in mild cognitive impairment and dementia. This research was performed

by the Cochrane Handbook for Systematic Reviews and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Higgins et al., 2019; Page et al., 2021). It was registered in PROSPERO (registration number: CRD42023417510).

2.1 | Core questions

The selection criteria of the articles included in this meta-analysis were examined using the participants, intervention, comparison, and outcome (PICO) framework.

2.1.1 | Participants (P)

Selection criteria for participants were diagnosis as mild cognitive impairment and dementia by a specialist or according to the standardized diagnostic criteria for inclusion (Albert et al., 2011).

2.1.2 | Intervention (I)

Digital technology interventions for improved cognitive function in mild cognitive impairment and dementia were selected. These interventions needed to be software developed to improve cognitive functions and installed and used in general-purpose hardware such as computers, mobile devices, and head-mounted displays. Those including software with other purposes (general health management, Wii sports games, etc.) were excluded.

2.1.3 | Comparison (C)

The comparison target was a group that did not undergo digital technology intervention and included no-treatment control, treatment as usual, standard care, cognitive behavioral therapy, and placebo groups.

2.1.4 | Outcome (O)

Cognitive function was measured before and after the digital technology intervention in participants with mild cognitive impairment or dementia. Outcomes included global cognitive function and neuropsychological characteristics, and the mean, standard error values, and standard deviation were used for analysis.

2.1.5 | Study design (S)

Only randomized controlled trials were selected to ensure a high level of evidence on the effects of digital technology interventions for

cognitive function improvement in participants with mild cognitive impairment or dementia.

2.2 | Search strategy and literature selection

Two authors conducted the search in April 2023 with no restrictions on the search period. Search databases were selected based on the Core, Standard, Ideal (COSI) model (Bidwell & Jensen, 2003). Literature searches were conducted in the PubMed, Cochrane Central Register of Controlled Trials, Cumulative Index to Nursing and Allied Health Literature (CINAHL), and Excerpta Medica Database (EMBASE) databases. Keyword selection and search included medical subject headings (MeSH) and life science term indices (EMBASE TREE; Emtree). All words similar to the search keywords were searched using the PubMed search function.

The keywords related to participants were "dementia" [MeSH], "dementias," "cognitive dysfunction" [MeSH], "cognitive," "dysfunction," and "mild cognitive impairment"; the intervention keywords comprised "digital," "software" [MeSH], "smartphone" [MeSH], "application," "app," "web," "virtual reality," "brain training," "brain exercise," "cognitive training," "cognitive exercise," "memory training," "cognitive training," "memory game," and "brain game." The keywords were searched by connecting them with "AND," and "OR." The research period was not limited to evaluate all relevant articles. The search strategies for each database are presented in Supporting Information S1: Material Table 1.

Inclusion criteria for the studies were: (1) intervention studies conducted using digital technology interventions for mild cognitive impairment and dementia, (2) randomized controlled trials, and (3) research published in English or Korean. Exclusion criteria were: (1) studies targeting participants without mild cognitive impairment or dementia, (2) studies targeting participants with comorbidities other than mild cognitive impairment and dementia, (3) pilot research that was not followed by the main study, (4) nonrandomized control experimental, qualitative (case study, descriptive study by interview), survey, literature review, animal experiment, and clinical trials of new drugs to treat mild cognitive and dementia, meta-analysis, and review, and (5) studies whose original text could not be found.

Two researchers (HJP and JYH) independently reviewed the studies for inclusion. Then, the researchers met to evaluate their individual decisions. In cases of disagreement, collective evaluation was completed, and a final decision was made through consensus.

2.3 | Data analysis

Two researchers (HJP and JYH) analyzed the articles included in the systematic literature review. Data were recorded for the year of publication, author, country of publication, participant age, number of samples, cognitive function screening tools, content of digital technology interventions, methods employed, number of interventions, cognitive function measurement tools, and outcome variables.

The heterogeneity and effect of the interventions in the selected research studies were examined using the RevMan 5.4 version of the Cochrane Collaboration. The main variable, heterogeneity, was assessed using a chi-square null hypothesis test. I^2 values of 0%, 30%–60%, and 75% indicated no, moderate, and high levels of heterogeneity, respectively (Wicks-Nelson & Israel, 2015). There was heterogeneity among the selected studies; therefore, they were analyzed using a random-effects model. A forest plot was used to confirm the direction of the effect and confidence interval (CI). The effect of the outcome value was represented by a standardized mean difference. The statistical significance level for the effect was set at 0.05, and the CI was set at 95%. Finally, a funnel plot was used to evaluate publication bias. Egger's regression test was conducted to examine the relationship between the standard error and effect to evaluate the significance of asymmetry.

2.4 | Quality assessment

The quality of the selected studies was evaluated using a revised tool to assess the risk of bias—the RoB 2.0 (The Cochrane Collaboration, England, London) (Higgins et al., 2019). The tool is organized into five domains: “randomization process,” “deviations from intended interventions,” “missing outcome data,” “measurement of the outcome,” and “selection of the reported result.” Evaluation results are recorded in three types: “low risk of bias,” “some concerns,” and “high risk of bias” (Sterne et al., 2019). The quality was assessed independently by both researchers (HJP and JYH). In instances of disagreement, the relevant studies were examined jointly until a consensus was reached, and RevMan (The Cochrane Collaboration, England, London) was used to present the evaluation results for risk (Page et al., 2021).

3 | RESULTS

3.1 | Study selection

The step-by-step literature selection process is depicted using a flowchart (Figure 1). The initial search retrieved 708 articles published from PubMed (424), Cochrane Library (52), CINAHL (16), and EMBASE (216). The searched studies were reviewed by titles, abstracts, and manuscripts based on the selection criteria. Twelve studies were finally selected.

3.2 | Quality evaluation

Methodological quality was evaluated for the 12 randomized controlled trials using RoB 2.0, and the risk of bias for randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result were evaluated. Nine studies had a low risk of bias because the randomization process and blinding were conducted appropriately

and three were described as randomly assigned; however, no clear method for randomization was reported. Overall, nine studies were rated as having a low risk of bias and three studies as having some concerns (Figure 2).

3.3 | Descriptive summary

The characteristics of the included studies are presented in Table 1. The countries where the studies were conducted were Korea (5, 41.7%), Taiwan (3, 25.0%), the UK (2, 16.7%), and France and Greece (1 each, 8.3%). Five studies (41.7%) were published between 2014 and 2019 and seven (58.3%) between 2020 and 2022.

The sample size was less than 50 in six studies (50.0%), and between 50 and 100 in six studies (50.0%), including the experimental and control groups. Regarding the gender of participants, the proportion of women was more than 70% of all participants in three studies (25.0%), 60%–70% in two studies (16.7%), and 50%–60% in three studies (25.0%). In two studies (16.7%), more than half of the participants were men. There were 11 studies (91.7%) that provided interventions for mild cognitive impairment (Han et al., 2017; Kang et al., 2021; Liao et al., 2019, 2020; Park, 2022a; 2022b; Robert et al., 2020; Savulich et al., 2017; Tarnanas et al., 2014; Thapa et al., 2020; Yang et al., 2019), and only 1 study (8.3%) that provided interventions for early-stage dementia (Rai et al., 2021).

All control groups received usual treatment instead of digital technology interventions. In the experimental group, seven studies (58.3%) employed virtual reality-based interventions, and five (41.7%) used memory training apps. Regarding the number of interventions, eight (3, 25.0%), 24 (2, 16.7%), 36 (3, 25.0%), 16 (1, 8.3%), 31 (1, 8.3%), and 48 sessions (1, 8.3%) were conducted. The duration of the interventions was 12 (4, 33.4%), eight (3, 25%), four (3, 25%), 11 (1, 8.3%), and 20 (1, 8.3%) weeks. Regarding the duration per session, six studies (50%) were conducted for less than 60 min and five (41.7%) for more than 60 min. In the selected 12 studies, global cognitive function, and neuropsychological characteristics were measured as the target outcomes of digital technology intervention. Regarding the time point of follow-up observation of the effects after providing the intervention, 10 studies (83.3%) measured the effects immediately after the intervention (Han et al., 2017; Kang et al., 2021; Liao et al., 2019, 2020; Park, 2022a; Park, 2022b; Savulich et al., 2017; Tarnanas et al., 2014; Thapa et al., 2020; Yang et al., 2019), two studies (16.7%) observed the effects within 12 weeks (Han et al., 2017; Rai et al., 2021), and two studies (16.7%) observed the effects more than 12 weeks later (Robert et al., 2020; Yang et al., 2019).

Of the 12 studies, six studies (50.0%) provided a control group with usual care that included physical or cognitive components (Han et al., 2017; Kang et al., 2021; Liao et al., 2019, 2020; Thapa et al., 2020; Yang et al., 2019), and six studies (50.0%) were no-treatment control studies (Park, 2022a; Park, 2022b; Rai et al., 2021; Robert et al., 2020; Savulich et al., 2017; Tarnanas et al., 2014).

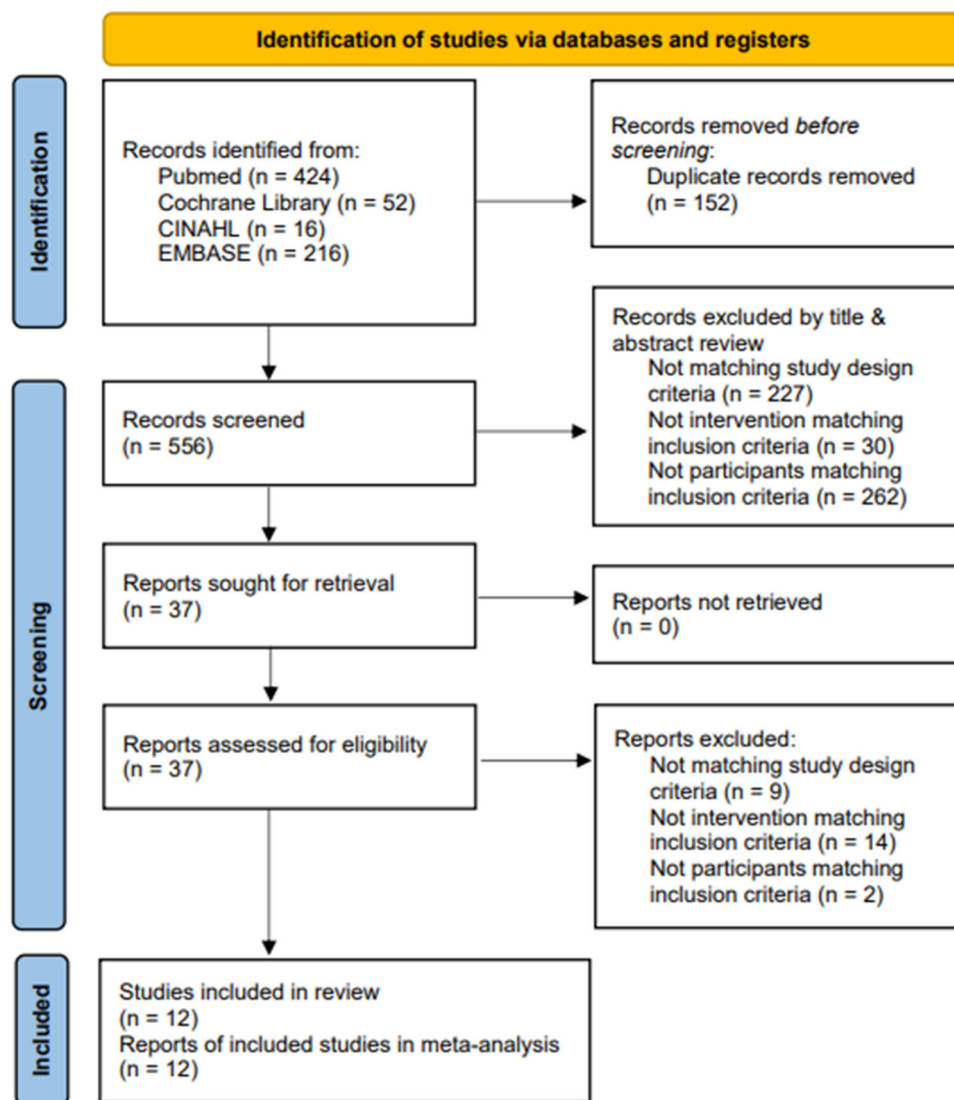


FIGURE 1 Flow diagram of study selection.

3.4 | Effect of digital technology intervention

A meta-analysis was conducted to analyze the effects of digital technology interventions on global cognitive function in mild cognitive impairment and dementia (Figure 3). Neuropsychological characteristics were classified into attention, visuospatial perception, memory, and frontal executive function. The effect to the digital technology interventions for cognitive function was calculated using the results immediately following the intervention in each study.

3.4.1 | Effect on global cognitive function

In 10 articles reporting the global cognitive function of the experimental group and control groups, the homogeneity test indicated $Q(\chi^2) = 22.35$, $df = 9$ ($p = 0.008$) and $I^2 = 60\%$. The standardized mean difference was 0.31 (95% CI: 0.04–0.57), which confirmed

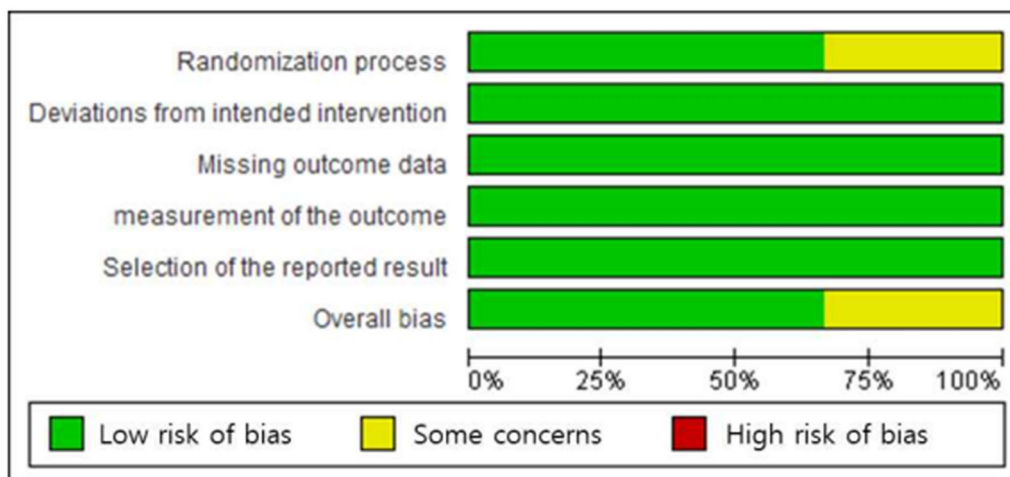
an effect on improving cognitive function, and there was a statistically significant difference in the global cognitive function between the experimental and control groups ($Z = 2.25$, $p = 0.02$).

Among the 10 studies that reported global cognitive function, excluding one study including early-stage dementia, thus analyzing nine studies targeting only mild cognitive impairment, $Q(\chi^2) = 15.33$, $df = 8$ ($p = 0.05$), and $I^2 = 48\%$. The standardized mean difference was 0.38 (95% CI: 0.13–0.63), which confirmed an effect on improving cognitive function, and there was a statistically significant difference between the experimental and control groups ($Z = 3.02$, $p = 0.003$).

3.4.2 | Effect on neuropsychological characteristics

Analysis of three articles reporting the effect of digital technology interventions on attention, the homogeneity test showed Q

(a) Risk of bias graph



(b) Risk of bias summary

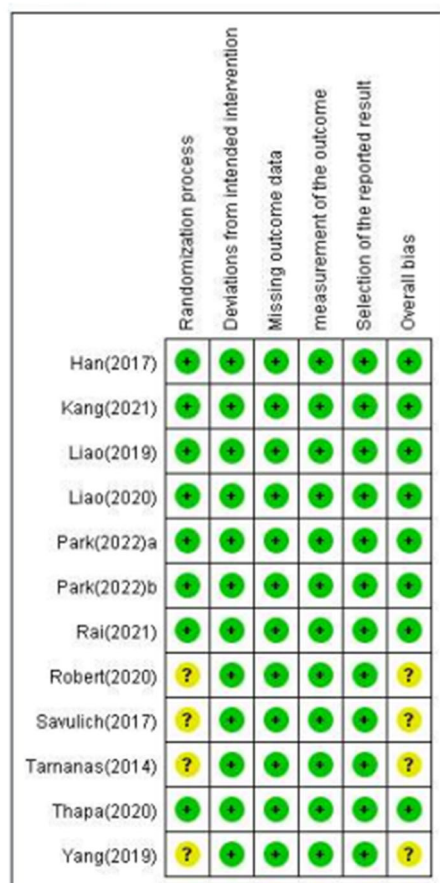


FIGURE 2 Assessment of risk of bias in included studies. (a) Risk of bias graph. (b) Risk of bias summary.

($\chi^2 = 12.21$, $df = 2$ ($p = 0.002$) and $I^2 = 84\%$). The standardized mean difference was 1.17 (95% CI: 0.36–1.97), and there was a significant difference in attention between the experimental and control groups ($Z = 2.85$, $p = 0.004$).

Three studies suggested the impact of digital technology interventions on visuospatial perception. The homogeneity test score was Q ($\chi^2 = 4.63$, $df = 2$ ($p = 0.10$) and $I^2 = 57\%$). The standardized mean difference was 0.68 (95% CI: 0.19–1.17), and there was a

TABLE 1 Characteristics of included studies (N = 12).

First author (year), country	Setting	Sample size (n)		Mean age (SD)		Intervention	Number of sessions/ intervention duration (weeks)/session duration (min)	Cognitive function measurement variables (instrument)
		Intervention group (male/ female)	Control group (male/female)	Intervention group	Control group			
Han et al. (2017), South Korea	Dementia clinic	43 (23/20)	42	74.01		Ubiquitous Spaced Retrieval-based Memory Advancement and Rehabilitation Training (USMART)	8/4/30	- Global cognition function (MMSE) - Word List - Memory (WLMT) - Subjective memory complaints (SMCQ)
Kang et al. (2021), South Korea	Memory clinic in hospital	23 (6/17)	18 (6/12)	75.48	73.28	Cognitive Training in Fully Immersive Virtual Reality	8/4/20–30	- Global cognition function (K-MMSE, CDR) - Visuospatial function (ROCF) - Attention (TMT) - Language and related functions (BNT) - Verbal memory (SVLT) - Frontal executive function (TMT, Stroop)
Liao et al. (2019), Taiwan	Day care center	18 (7/11)	16 (4/12)	75.5	73.1	Virtual Reality-Based Physical and Cognitive Training	36/12/60	Executive function (TMT, SCWT)
Liao et al. (2020), Taiwan	Community center	18 (7/11)	16 (4/12)	75.5	73.1	Virtual reality-based physical and cognitive raining	36/12/60	- Global cognitive function (MoCA) - Executive function (EXIT-25) - Verbal memory (CVVLT)
Park (2022a), South Korea	Local senior center	28 (12/16)	28 (11/17)	71.93	72.04	Virtual reality-based spatial cognitive training	24/8/45	- Global cognitive function (WAIS) - Episodic memory (SVLT)
Park (2022b), South Korea	Local senior center	16 (9/7)	16 (6/10)	72.25	70.88	Virtual shopping training using the virtual supermarket application	16/8/-	Executive function (EFPT-K)
Rai et al. (2021), UK	Community- dwelling	31 (22/9)	30 (20/10)	74.03	71.81	Individual cognitive stimulation therapy application (iCST app)	31/11/30	Global cognitive function (ADAS-Cog)
Robert et al. (2020), France	Memory center	25 (15/10)	21 (7/14)	79.8	78.8	Web App for cognitive training (Memory motivation; MeMo)	48/12/30	- Global cognitive function (MMSE, IQCODE) - Episodic memory (TMT, Stroop, DSST, FAB)

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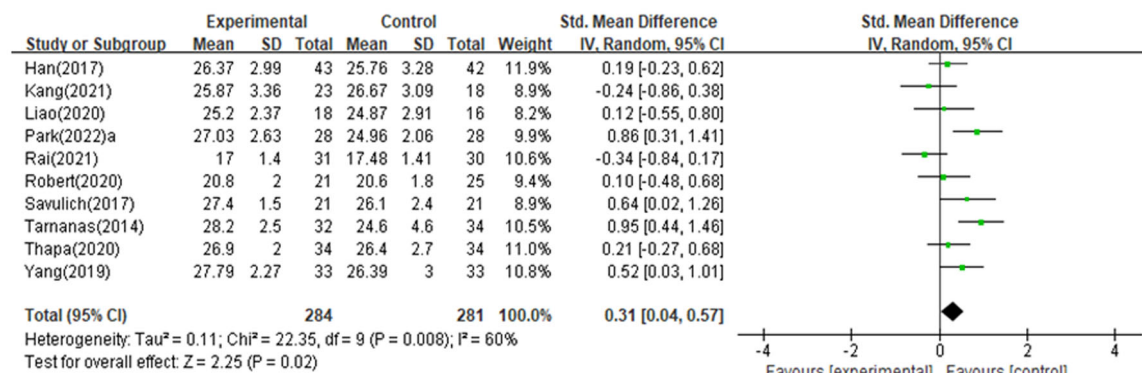
TABLE 1 (Continued)

First author (year), country	Setting	Sample size (n)		Mean age (SD)		Intervention	Number of sessions/ intervention duration (weeks)/session duration (min)	Cognitive function measurement variables (instrument)
		Intervention group (male/ female)	Control group (male/female)	Intervention group	Control group			
Savulich et al. (2017), UK	Memory clinic	21 (11/10)	21 (14/7)	75.2	76.9	Novel memory game on an iPad (Game show)	8/4/60	- Global cognitive function (MMSE) - Episodic memory (CANTAB)
Tammanas et al. (2014), Greece	Dementia clinic	32	34	70.5	70.9	Museum-based cognitive training using Virtual Reality, Augmented Reality and Web3D technologies in conjunction with database technology	~20/90	- Global cognitive function (MMSE) - Memory (RAVLT) - Praxis and visuospatial skills (ROCF) - Attention and executive function (Symbol Digit Modalities Test, TMT, Stroop)
Thapa et al. (2020), South Korea	Medical center	34 (6/28)	34 (10/24)	72.6	72.7	Immersive Virtual reality-based cognitive training	24/8/100	- Global cognitive function (MMSE, MoCA) - Cognitive function (TMT, DSST)
Yang et al. (2019), Taiwan	Retirement community	33 (8/25)	33 (6/27)	75.4	81.7	virtual interactive working memory training (CogniPlus)	36/12/45	- Global cognitive function (MMSE, MoCA) - Working memory (digit span) - Immediate memory (WMS-III) - Delayed memory (WMS-III) - Subjective memory complaints (MMQ)

Abbreviations: ADAS-Cog, Alzheimer's Disease Assessment Scale-Cognition; BNT, Boston Naming Test; BVM-T-R, Brief Visuospatial Memory Test-Revised; CANTAB, Cambridge Neuropsychological Test Automated Battery; CDR, Clinical Dementia Rating; CVLT, Chinese Version Verbal Learning Test; DSST, Digit Symbol Substitution Test; EXIT, Executive Intelligence Test; FAB, Frontal Assessment Battery; IQCODE, Informant Questionnaire on Cognitive Decline in the Elderly; MMQ, Multifactorial Memory Questionnaire; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; RAVLT, Rey Auditory Verbal Learning Test; ROCF, Rey-Osterrieth Complex Figure; SCWT, Stroop Color and Word Test; SMCQ, Subjective Memory Complaint Questionnaire; SVLT, Seoul Verbal Learning Test; TMT, Trail Making Test; WAIS, Wechsler Adult Intelligence Scale; WLMT, the Word List Memory Test; WMS-III, Wechsler Memory Scale-Third Edition.

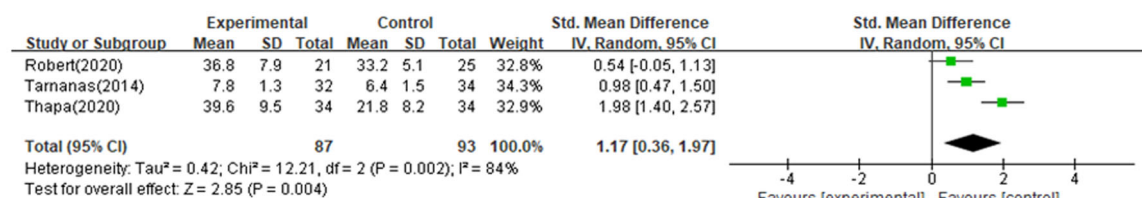
Analysis 1.1 Intervention vs Control, Outcome 1: Global cognition

1.1.1 Global cognition

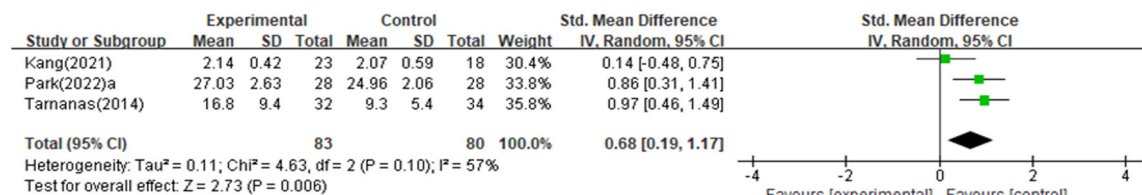


Analysis 1.2 Intervention vs Control, Outcome 2: Neuropsychological characteristics

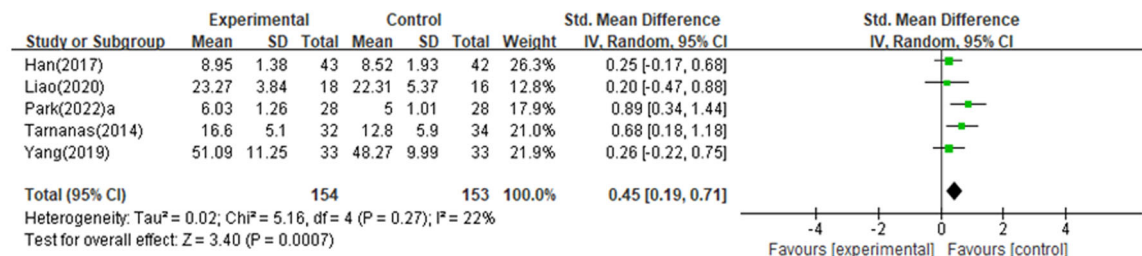
1.2.1 Attention



1.2.2 Visuospatial perception



1.2.3 Memory



1.2.4 Frontal executive function

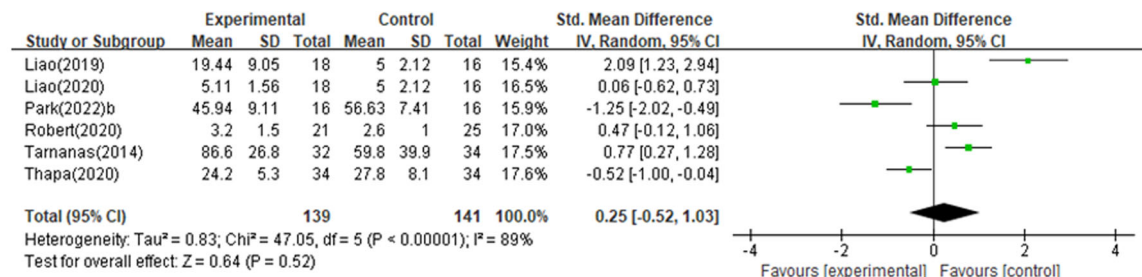


FIGURE 3 Forest plots of the effects of digital technology interventions. CI, confidence interval; SD, standard deviation.

significant difference in visuospatial perception between the experimental and control groups ($Z = 2.73$, $p = 0.006$).

Five studies evaluated the effect of digital technology interventions on memory. The homogeneity test indicated $Q (\chi^2) = 5.16$, $df = 4$ ($p = 0.27$) and $I^2 = 22\%$. The standardized mean difference was 0.45 (95% CI: 0.19–0.71), and there was a significant difference in memory between the experimental and control groups ($Z = 3.40$, $p = 0.0007$).

Six studies reported the frontal executive function of experimental and control groups. The homogeneity test showed $Q (\chi^2) = 47.05$, $df = 5$ ($p < 0.00001$) and $I^2 = 89\%$. The standardized mean difference was 0.25 (95% CI: -0.52, 1.03), and there was no statistically significant difference in frontal executive function between the experimental and control groups ($Z = 0.64$, $p = 0.52$).

3.5 | Publication bias

Publication bias was assessed by visual and intuitive funnel plot. Each study was distributed around the effect estimate, and the shape of the graph was symmetrical with respect to the centerline, which did not digress significantly from right-left symmetry. Therefore, we concluded there was no publication bias (Figure 4). In addition, Egger's regression test confirmed the significance as follows: global cognitive function ($p = 0.487$), attention ($p = 0.236$), and visuospatial

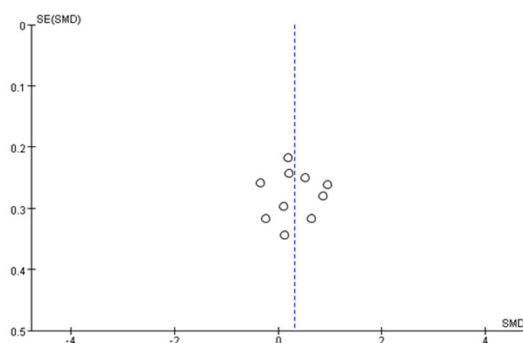
perception ($p = 0.252$), memory ($p = 0.325$), and frontal executive function ($p = 0.287$) had no publication bias.

4 | DISCUSSION

In this systematic review and meta-analysis, we identified and evaluated digital technology interventions for improving cognitive function in mild cognitive impairment and dementia through 12 published randomized controlled trials. Despite significant variations across the studies, the meta-analysis revealed that digital technology interventions significantly increased global cognitive function in mild cognitive impairment and dementia. In addition, it significantly improved attention, visuospatial perception, and memory, but not executive function.

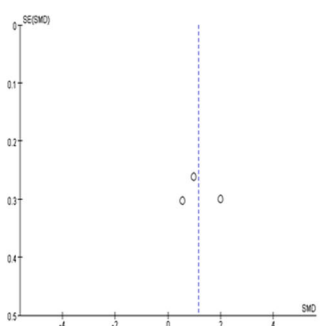
Consistent with the recent proliferation of digital technologies in social and healthcare interventions, all 12 randomized controlled trials included in this review were published after 2014. Only one study provided intervention for early-stage dementia, and the rest targeted mild cognitive impairment. This appears to reflect the importance of therapeutic intervention in mild cognitive impairment and early dementia stages in reducing the prevalence of dementia. In the Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability (FINGER), a large, long-term randomized

(a) Global cognitive function

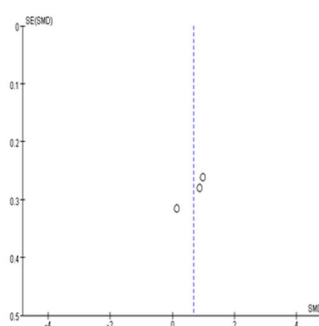


(b) Neuropsychological characteristics

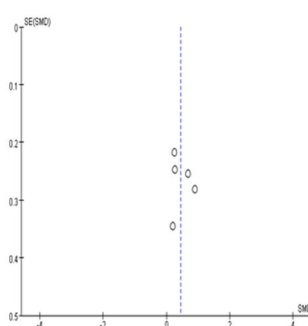
a. Attention



b. Visuospatial perception



c. Memory



d. Executive function

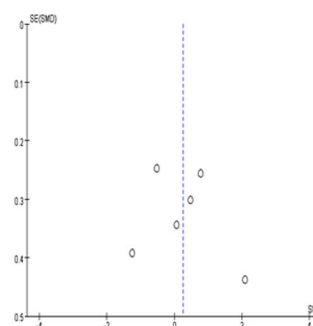


FIGURE 4 Funnel plots. (a) Global cognitive function. (b) Neuropsychological characteristics.

controlled trial, cognitive function was improved or maintained when multidomain intervention was provided to older adults at high risk of dementia (Ngandu et al., 2015). Additionally, the ACTIVE study in the United States provided cognitively healthy participants with specific training on memory, reasoning, and processing speed, and demonstrated a significantly lower incidence of dementia in the intervention group at five and 10-year follow-up (Jobe et al., 2001). On the other hand, the results of clinical interventions provided in the dementia stage of Alzheimer's disease were mostly disappointing, suggesting that the participants had already exceeded the optimal treatment period for intervention (Schneider et al., 2014). In this way, evidence suggests that intervention to maintain and improve cognitive function must be carried out before the complete transition to dementia. In this study, most of the studies provided interventions for mild cognitive impairment, so it was not possible to compare the effect size of the intervention depending on the level of cognitive function. Future research may more clearly elucidate these condition-specific effects of a digital technology intervention.

Digital technology interventions for mild cognitive impairment and dementia varied widely in terms of content, composition, duration, and frequency. Digital technologies such as mobile and tablet applications and virtual reality were used for activity engagement interventions. For mobile and tablet applications, interventions have applied cognitive training, cognitive rehabilitation, and memory training. Virtual reality interventions were provided through cognitive training, shopping training, and virtual museums. A detailed analysis of effective digital technology interventions was not possible because of the differences in intervention methods, and frequency and time of provision. In future, large-scale studies of various digital technology intervention methods need to be conducted.

The results of this study confirmed that digital technology interventions had a significant effect on the global cognitive function of mild cognitive impairment and dementia. This is consistent with the results of improvement in overall cognitive function in mild cognitive impairment in a meta-analysis that examined the effect of computerized cognitive training (Hill et al., 2017). Additionally, these results are similar to the results of a meta-analysis that examined the effects of virtual reality-based interventions (Zhong et al., 2021). The ultimate goal of interventions for cognitive function for mild cognitive impairment and dementia is to improve brain reorganization and cognitive function through direct learning and training. Through this, physical, cognitive, and psychosocial functions improve, leading to an independent life and enabling social activities (Lezak et al., 2004). For this, a standardized technique that reflects specific cognitive abilities is adopted, and the process of sequential and repetitive practice is applied to various forms of intervention (Bahar-Fuchs et al., 2013; Ballard et al., 2011); digital technology intervention for mild cognitive impairment and dementia could also be effective in these purposes. Therefore, it will be helpful to utilize digital technology intervention as a treatment method for improving the cognitive function in mild cognitive impairment or dementia and slowing down the progression to dementia.

Long-term effects analysis is needed to understand the cycle of delivering digital technology interventions. In this study, only short-term effects were confirmed as the effects were measured immediately or within 12 weeks after the intervention was provided. The two studies that measured the effect of the intervention at three (Yang et al., 2019) and six (Robert et al., 2020) months had differences in the measured outcome variables, making it impossible to compare effect sizes. According to the results of the two studies, the effect partially persisted or decreased, but future research should be conducted to examine the long-term effect in maintaining and improving cognitive function.

Mild cognitive impairment and dementia are diagnosed by dividing the cognitive domains into attention, executive function, memory, visuospatial perception ability, and language ability, according to standardized clinical evaluations (Sachdev et al., 2014). Therefore, in this study, the effects of digital technology interventions on neuropsychological characteristics were analyzed in each area: attention and concentration, frontal executive function, memory, and visuospatial perception. Among the cognitive domains, language ability was measured in only one of the 12 selected studies; therefore, the effect could not be analyzed. More intervention studies suggesting the effect on language ability and determining the effect should be conducted in the future.

Our analysis indicated that the neuropsychological characteristics intervention most affected by the intervention was attention and concentration, followed by visuospatial perception and memory. These findings are consistent with the results of computer-based cognitive training, which showed an improvement in memory, based on a meta-analysis (Hill et al., 2017). In addition, semi-immersive virtual reality-based cognitive training improved cognitive function in patients who had a stroke, and improved attention, concentration, and spatial perception ability (De Luca et al., 2018). In contrast, a meta-analysis revealed no significant effect of virtual reality cognitive training on memory or attention in adults with mild cognitive impairment. Disparate findings in previous studies may be due to lack of inclusion of virtual reality interventions using devices, such as Xbox and Nintendo, and software for improving cognitive function in dementia. Further, the sample sizes were not large; therefore, the results may be unstable. To prepare a strong evidence base for digital technology intervention as a treatment for dementia, it is necessary to accumulate more rigorous and elaborately designed intervention studies to draw strong conclusions about their effect on neuropsychological characteristics.

Our results indicate that digital technology interventions may not affect frontal executive function. Frontal executive function is responsible for strategically planning for successful plan implementation. In other words, executive function refers to the ability to create performance steps to reach a goal, motivate each step to be achieved, and maintain control until the goal is reached (Mendez & Cummings, 2003). In older adults without cognitive impairment, motor training intervention was effective in improving executive function (Chen et al., 2020), and mobile-based cognitive intervention significantly improved executive function (Ha & Park, 2023). Given

that interventions are effective in improving executive function in older adults with normal cognitive function, it is necessary for cognitive intervention to occur before progressing to moderate or severe dementia. In addition, digital technology interventions should be supplemented to improve executive function, a higher-level cognitive system, beyond the immediate follow-up of simple instructions. To do so, the functions of daily life at a more complex level should be integrated in a goal-oriented manner.

Based on the above results, digital technology interventions for mild cognitive impairment and dementia can help maintain cognitive function or delay the transition to dementia and can be used as a digital therapeutic for dementia in the future. However, our results on the effect of digital technology interventions on the cognitive domain suggest it is necessary to seek ways to improve frontal executive function. Thus, not only the improvement of global cognitive function but also that of cognitive function by detailed functional area needs to be prioritized.

Most of the studies examined in this paper were conducted with a small sample size, and there was heterogeneity regarding the participants, intervention methods, and outcome measures. This resulted in limitations such as high heterogeneity and wide confidence intervals. The measures reported in the selected papers were standardized tools, but the tools used in each study varied, so there are limitations in generalizing the results. Therefore, large-scale clinical studies with additional analysis of effective intervention methods should be conducted. Additionally, a long-term follow-up study should be conducted to confirm the maintenance of digital technology intervention effects. Nevertheless, this study identified the characteristics of digital technology interventions on mild cognitive impairment and dementia, analyzed their effect on global cognitive function, and compared and analyzed their effect by cognitive area to improve their effects in mild cognitive impairment and dementia. The study is meaningful in that it provides a basis for digital therapeutic development for cognitive dysfunction.

5 | CONCLUSIONS

In this study, digital technology intervention appeared to have a significant effect on improving global cognitive function for mild cognitive impairment and dementia. Moreover, digital technology interventions could be considered a functional approach for managing the neuropsychological characteristics of attention, visuospatial perception, and memory. However, the effect of digital technology intervention on frontal executive function is inconclusive, as there is insufficient evidence to evaluate its effects. Therefore, future clinical studies with strict controls and studies that analyze the maintenance of effects will help establish the basis for digital technology intervention. The results of this study are expected to serve as a stepping-stone for developing and implementing digital therapeutics to maintain and improve cognitive function in mild cognitive impairment and dementia.

AUTHOR CONTRIBUTIONS

Hyojin Park: Study design; conception; literature screening and data extraction; quality appraisal of selected studies; data analysis and interpretation; visualization; writing—original draft; writing—review and editing. **Juyoung Ha:** Supervision; literature screening and data extraction; quality appraisal of selected studies; data analysis and interpretation; writing—review and editing. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

All the authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request. All data generated or analyzed during this study are included in the published article. For further clarifications, the authors can be contacted at hyojin@pusan.ac.kr.

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SUPPORTING INFORMATION

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